

SYSTEM REQUIREMENT SPECIFICATION DOCUMENT

SUBMITTED BY

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Chapter 1 – Introduction

1.1 Project Background

The concept for the Smart Tourism Platform originated from the growing challenges faced by travelers when planning trips across Sri Lanka. Tourists often rely on multiple disconnected platforms and manual processes to arrange accommodation, transportation, tourist guides, and destination information. This fragmented approach creates inconvenience, consumes time, and reduces the overall travel experience. Recognizing these issues, the Smart Tourism Platform was proposed as a centralized digital solution to simplify and modernize tourism management within Sri Lanka.

Currently, tourists use separate applications and websites for hotel reservations, transportation booking, destination exploration, and communication with tourist guides. These systems operate independently, causing inefficiencies and making trip planning more complicated. Travelers frequently face difficulties such as inconsistent information, lack of coordination between services, and limited access to hidden or less popular destinations. Additionally, local tourism service providers often struggle to reach a wider audience due to the absence of an integrated tourism ecosystem.

Traditional tourism management methods, including manual bookings and physical inquiries, are increasingly outdated in today's digital environment. Modern travelers expect convenient, real-time, and personalized services through a single online platform. However, existing systems fail to provide a seamless experience that integrates all tourism-related services together.

The Smart Tourism Platform aims to bridge this gap by offering an all-in-one solution that combines hotel booking, transport services, tourist guide management, and destination exploration within a centralized platform. The system is designed to enhance user convenience, improve communication between tourists and service providers, and promote both famous and hidden tourist destinations in Sri Lanka.

1.2 Problem Definition

The current tourism management systems in Sri Lanka are fragmented, inefficient, and lack integration. Travelers are required to use multiple independent platforms and manual methods to organize different aspects of their trips, including accommodation, transportation, tourist guides, and destination planning. This disconnected process complicates travel management and creates unnecessary difficulties for both tourists and tourism service providers.

The absence of a centralized tourism platform limits accessibility to reliable information and reduces the visibility of many local tourism businesses and hidden destinations. Furthermore,

tourists often experience delays, inconsistent booking processes, communication barriers, and difficulties in coordinating travel services effectively.

Traditional tourism practices fail to satisfy the expectations of modern travelers who demand fast, digital, and user-friendly experiences. Therefore, there is a strong need for a smart, integrated tourism platform that simplifies travel planning, improves service accessibility, and enhances the overall tourism experience in Sri Lanka.

1.3 Aims and Objectives

Aim:

The primary aim of the Smart Tourism Platform is to develop a centralized and efficient tourism management system that simplifies trip planning and enhances the travel experience in Sri Lanka.

Objectives:

- **Streamline Tourism Services:**

The platform will integrate hotel booking, transport services, tourist guide management, and destination exploration into a single system, eliminating the need for multiple disconnected platforms.

- **Enhance User Experience:**

The system will provide an intuitive and user-friendly interface that allows travelers to easily search, book, and manage tourism-related services in real time.

- **Promote Tourism in Sri Lanka:**

The platform will showcase both famous and hidden destinations, helping tourists discover new places while supporting local tourism businesses and communities.

- **Improve Communication and Coordination:**

Integrated communication features and notifications will allow tourists, hotel providers, transport providers, and tourist guides to coordinate efficiently.

- **Support Service Providers:**

Hotel owners, transport providers, and tourist guides will have dedicated dashboards to manage profiles, services, bookings, and availability effectively.

- **Provide Secure and Reliable Services:**

The system will implement secure authentication and booking management features to ensure safe and reliable transactions for all users.

By achieving these objectives, the Smart Tourism Platform aims to modernize tourism management in Sri Lanka by providing a convenient, centralized, and technology-driven solution for travelers and service providers.

1.4 Scope with Clear Boundaries

The scope of the Smart Tourism Platform is to develop a centralized web-based system that supports tourism-related services and improves travel planning within Sri Lanka. The system focuses on integrating essential tourism functionalities into a single platform while maintaining a user-friendly experience.

Features to be Implemented:

- **User Registration and Authentication:**
 - Users can register and log in securely based on their roles.
 - JWT-based authentication and role-based authorization will be implemented for secure system access.
- **Hotel Booking Management:**
 - Hotel providers can register and manage hotel profiles.
 - Providers can add rooms, update room availability, manage pricing, and handle bookings.
 - Users can search and book hotels through the platform.
- **Transport Service Management:**
 - Transport providers can register and manage transport services.
 - Users can book transportation services for their trips.
- **Tourist Guide Management:**
 - Tourist guides can create profiles and manage their services.
 - Users can browse and hire tourist guides based on preferences.
- **Destination Exploration:**
 - The platform will provide detailed information about famous and hidden tourist destinations in Sri Lanka.
 - Users can search destinations and view travel-related information.

- **Package Booking System:**
 - Users will be able to combine hotels, transport, and tourist guide services into a single travel package booking.
- **Admin Management:**
 - Administrators will manage users, service providers, approvals, destinations, and overall platform operations.
- **Notifications and Updates:**
 - Users and providers will receive notifications regarding bookings, approvals, and service updates.
- **Mobile Application Support:**
 - A dedicated mobile application will be developed to provide better accessibility for users.
- **Payment Gateway Integration:**
 - The system will support secure online payments for bookings and service reservations.

Features Out of Scope:

1. **Advanced AI Recommendation System:**

Although basic destination suggestions may be provided, advanced AI-powered personalized travel recommendations are not included in this phase.
2. **Social Media Integration:**

Direct integration with social media platforms for automatic content sharing is outside the scope of this project.
3. **Multi-Country Tourism Support:**

The platform is specifically designed for tourism services within Sri Lanka and will not support international tourism services in this phase.

1.5 Organization of the Dissertation

This dissertation is organized into the following chapters to provide a complete understanding of the Smart Tourism Platform project:

Chapter 01: Introduction

This chapter introduces the Smart Tourism Platform and explains the background and motivation behind the project. It discusses the current issues in tourism management systems, defines the problem statement, and presents the aims, objectives, and project scope. Additionally, the chapter outlines the overall structure of the dissertation.

Chapter 02: System Analysis and Requirements

This chapter focuses on analyzing the existing tourism management processes and identifying system requirements. It includes requirement gathering methods, feasibility analysis, UML diagrams, use case analysis, and Software Requirement Specifications (SRS). Functional and non-functional requirements are also discussed in detail.

Chapter 2 – System Analysis

2.1 Chapter Introduction

In this chapter, we present a comprehensive analysis of the Smart Tourism Platform for Sri Lanka, focusing on requirements gathering and analysis from the user's perspective. The chapter explores how software requirements were identified, including both functional and non-functional requirements, and explains the core features required to support a unified tourism ecosystem.

Additionally, we examine the system architecture and key design considerations necessary to develop a centralized platform that integrates hotel booking, transport services, tour guide services, and information about both famous and hidden destinations in Sri Lanka. Through this analysis, we establish a strong foundation for addressing the fragmentation in current tourism services and enhancing the overall travel experience for users.

2.2 Fact Gathering Techniques Used

To ensure a thorough understanding of user expectations and system requirements, two primary fact-gathering techniques were employed during the development of the Smart Tourism Platform. These techniques included:

- **Survey Analysis:**

A structured survey was conducted among potential users, including local travelers and tourists, to understand their travel behavior, booking preferences, and challenges faced when planning trips in Sri Lanka. The survey helped identify common issues such as the need to use multiple platforms for booking hotels, transport, and activities, as well as difficulties in discovering lesser-known tourist destinations. The responses highlighted a strong demand for a centralized platform that integrates all travel-related services.

- **Competitive Analysis:**

An in-depth analysis of existing tourism and travel platforms was conducted to evaluate their features, usability, and limitations. This study included both local and international platforms. The analysis revealed gaps such as lack of integration between services, limited personalization, and insufficient support for discovering hidden tourist attractions. These findings were used to guide the design of the Smart Tourism Platform to ensure improved functionality and user experience.

By applying these fact-gathering techniques, the development team gained a clear understanding of user needs and market gaps. The insights gathered played a crucial role in shaping the Smart Tourism Platform into an efficient and user-friendly system.

2.2.1 Survey Questions and Responses

Below are the key questions included in the survey, along with a summary of the responses collected from participants.

Questions for Users / Tourists

1. How often do you travel for tourism?

Answer:

Most respondents travel at least once a year, with a considerable portion traveling two to three times annually or more frequently. Only a small number were first-time travelers, indicating that the majority have prior travel experience and an ongoing need for better planning tools.

2. What type of traveler are you?

Answer:

The majority of respondents travel in groups, followed by those who travel with family. Very few travel solo or as couples. This suggests the platform should focus on features that support group coordination, shared planning, and family-friendly travel arrangements.

3. How do you currently plan your trips?

Answer:

Most respondents rely heavily on recommendations from friends and family, while others use social media and online platforms. Very few use formal travel agencies. This reveals a gap in structured planning tools and a strong opportunity for a dedicated travel platform.

4. How much time do you usually spend planning a trip?

Answer:

The majority spend between one to three days planning a trip, while others spend over a week. Very few plan in under a day. This highlights that trip planning is time-consuming and needs simplification.

5. What is the most difficult part of planning a trip?

Answer:

Budgeting is the most difficult aspect, followed by choosing destinations, arranging transport, finding accommodation, and safety concerns. This shows users struggle across multiple planning stages.

6. What influences your travel decisions the most?

Answer:

Friends and family recommendations are the strongest influence, followed by social media, price, and reviews/ratings. Travel blogs have less influence. This indicates strong value for community-driven features.

7. What is your average travel budget?

Answer:

Most respondents spend less than LKR 15,000 per trip. A smaller portion falls in mid-range budgets, and very few travel with high budgets. This shows the importance of affordable and budget-friendly planning tools.

8. What do you value most when traveling?

Answer:

Comfort and unique experiences are the top priorities, followed by safety and low cost. This shows users want both affordability and meaningful experiences.

9. What problems did you face when traveling?

Answer:

Common issues include poor planning, language barriers, hidden costs, lack of destination information, safety concerns, and unreliable internet. Some also reported issues with navigation, food, and medical access.

10. How satisfied are you with current travel planning methods?

Answer:

Most respondents feel neutral, indicating current tools are functional but not optimal. Very few are highly satisfied, showing a clear opportunity for improvement.

11. Would you use a platform that creates personalized travel plans based on budget and interests?

Answer:

An overwhelming majority responded positively, showing strong demand for a smart, personalized travel planning solution.

12. What features would you like in a travel platform?

Answer:

Top requested features include:

- Budget planner
- Local experience suggestions
- Safety alerts
- Transport guidance
- Reviews and ratings

Conclusion of Survey Analysis

The survey results clearly show that tourists face major challenges due to the lack of a centralized and intelligent travel planning system. Most users rely on informal recommendations, spend significant time planning, and face issues such as budgeting difficulties, hidden costs, and lack of reliable information.

There is strong demand for a **Smart Travel Planning Platform** that integrates budget management, destination discovery, transport coordination, accommodation booking, and safety information into a single system. The findings strongly support the development of an affordable, user-friendly, and personalized travel solution.

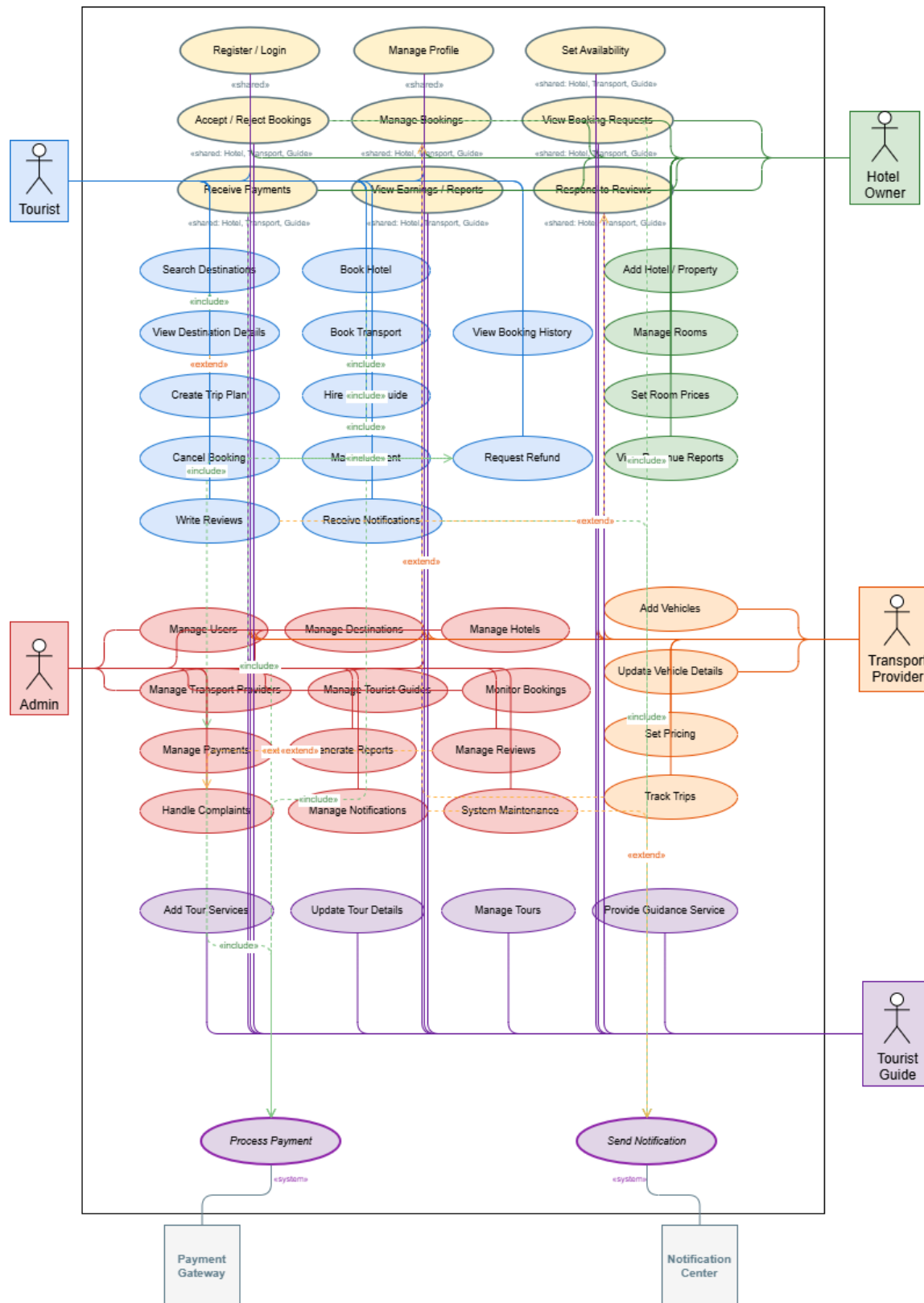
2.2.2 Competitive Analysis Report

Feature	Booking.com	Airbnb	Tripadvisor
Ease of Navigation	High	High	Medium
UI Simplicity	Medium	High	Medium
Booking Experience	Excellent	Good	Medium
Search Experience	Excellent	Good	Good
Mobile Friendliness	Good	Excellent	Good
Reviews & Trust	High	High	Excellent
Pricing Transparency	High	Medium	Medium
Personalization	Medium	High	Medium

Competitive Analysis Summary

A comparative analysis of Booking.com, Airbnb, and Tripadvisor shows that each platform has different strengths. Booking.com performs best in search, booking experience, and pricing transparency, while Airbnb excels in UI simplicity, mobile friendliness, and personalization. Tripadvisor stands out in reviews and trust due to its strong user-generated content. However, no platform performs strongly in all areas, highlighting the need for a unified tourism system that combines the best features of each while improving overall user experience.

2.3 Use case diagram



2.4 Use case description

Use Case ID	UC-01
Use Case Name	Register / Login
Actor(s)	Tourist, Hotel Owner, Transport Provider, Tourist Guide (Common to all actors – flows described for Tourist; other actors follow the same pattern with their respective dashboards)
Preconditions	User has internet access and valid credentials for login
Postconditions – Success	User account is created or user is successfully logged into the system
Postconditions – Failure	Registration/login fails; error message displayed
Main Flow	1. User opens the system. 2. Selects Register or Login. 3. Enters required details. 4. System validates information. 5. System creates account or authenticates user. 6. User is redirected to their role dashboard.
Alternative Flow 1	At step 4: If credentials are invalid, system displays error message.
Alternative Flow 2	At step 3: If email already exists during registration, system prompts user to login instead.
Business Rules	BR-01: Email must be unique. BR-02: Password must meet minimum security requirements.

Use Case ID	UC-02
Use Case Name	Manage Profile
Actor(s)	Tourist, Hotel Owner, Transport Provider, Tourist Guide (Common to all actors)
Preconditions	User is logged into the system
Postconditions – Success	Profile information is updated successfully
Postconditions – Failure	Profile remains unchanged; validation error displayed
Main Flow	1. User opens profile settings. 2. System displays current profile information.

	3. User edits details. 4. User submits changes. 5. System validates and saves updates.
Alternative Flow 1	At step 5: If required fields are empty or invalid, system displays validation errors.
Alternative Flow 2	User cancels editing; system returns to dashboard without changes.
Business Rules	BR-03: Contact details (phone, email) must follow valid format requirements.

Use Case ID	UC-03
Use Case Name	Set Availability
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Provider is logged in and has registered services/properties/vehicles
Postconditions – Success	Availability schedule is updated successfully
Postconditions – Failure	Availability update fails
Main Flow	1. Provider opens availability settings. 2. Selects service, vehicle, or room. 3. Chooses available dates and time slots. 4. System validates information. 5. Availability schedule is saved.
Alternative Flow 1	At step 4: If selected slots overlap or exceed capacity, system displays validation error.
Alternative Flow 2	Provider cancels changes before saving.
Business Rules	BR-04: Resources cannot be assigned to overlapping reservations. Availability cannot exceed total capacity.

Use Case ID	UC-04
Use Case Name	Accept / Reject Bookings
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Pending booking requests exist in the system

Postconditions – Success	Booking request status is updated
Postconditions – Failure	Booking request remains pending
Main Flow	1. Provider views booking requests. 2. Selects a request. 3. Reviews booking details. 4. Accepts or rejects the booking. 5. System updates booking status and notifies the tourist.
Alternative Flow 1	At step 3: If the resource (room/vehicle/guide slot) becomes unavailable, system automatically rejects the request.
Alternative Flow 2	Provider delays decision; request remains pending.
Business Rules	BR-05: Booking decisions must be recorded with timestamps.

Use Case ID	UC-05
Use Case Name	Manage Bookings
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Provider is logged into the system
Postconditions – Success	Booking details are updated successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Provider opens booking management section. 2. System displays current bookings. 3. Provider updates booking details or status. 4. System validates changes. 5. Updated booking information is saved.
Alternative Flow 1	At step 4: If update conflicts with another booking, system rejects changes.
Alternative Flow 2	Provider cancels modifications before saving.
Business Rules	BR-06: Double booking of resources is prohibited.

Use Case ID	UC-06
Use Case Name	View Booking Requests
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Pending booking requests are available in the system
Postconditions – Success	Booking requests are displayed
Postconditions – Failure	Error message displayed
Main Flow	1. Provider selects booking requests option. 2. System retrieves pending requests. 3. System displays request details.
Alternative Flow 1	At step 2: If no requests exist, system displays empty request list.
Alternative Flow 2	Provider filters requests by date, room type, vehicle type, or destination.
Business Rules	BR-07: Requests must be displayed in chronological order.

Use Case ID	UC-07
Use Case Name	Receive Payments
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide, Payment Gateway (Common to service providers)
Preconditions	Tourist payment has been completed
Postconditions – Success	Payment is credited to provider account successfully
Postconditions – Failure	Payment transaction fails
Main Flow	1. Tourist completes payment. 2. Payment gateway processes transaction. 3. System verifies and confirms payment. 4. Provider receives payment notification.
Alternative Flow 1	At step 2: If payment fails, system displays transaction error.
Alternative Flow 2	Payment processing is delayed; system marks transaction as pending.
Business Rules	BR-08: All payment transactions must be securely processed.

Use Case ID	UC-08
Use Case Name	View Earnings / Reports
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Provider has completed bookings with associated transactions
Postconditions – Success	Earnings reports are generated and displayed successfully
Postconditions – Failure	Report generation fails
Main Flow	1. Provider opens reports section. 2. Selects report period. 3. System retrieves transaction data. 4. System generates report. 5. Earnings report is displayed.
Alternative Flow 1	At step 3: If no data exists for the selected period, system displays empty report message.
Alternative Flow 2	Provider exports report for offline use.
Business Rules	BR-09: Reports must include only completed/confirmed transactions.

Use Case ID	UC-09
Use Case Name	Respond to Reviews
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Tourists have submitted reviews for completed bookings
Postconditions – Success	Response is published successfully
Postconditions – Failure	Response submission fails
Main Flow	1. Provider views customer reviews. 2. Selects a review. 3. Writes response. 4. Submits response. 5. System publishes response.
Alternative Flow 1	At step 4: If response violates content policies, system rejects it.
Alternative Flow 2	Provider edits response before submission.

Business Rules	BR-10: Responses must comply with platform review guidelines and content policies.
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Use Case ID	UC-10
Use Case Name	Search Destinations
Actor(s)	Tourist
Preconditions	Tourist is logged into the system
Postconditions – Success	Matching destinations are displayed
Postconditions – Failure	No destinations found message displayed
Main Flow	1. Tourist enters search criteria. 2. System processes the request. 3. System retrieves matching destinations. 4. Results are displayed to the tourist.
Alternative Flow 1	At step 3: If no matches are found, system displays "No destinations available".
Alternative Flow 2	Tourist clears filters and performs a new search.
Business Rules	BR-11: Search results must match selected filters and keywords.

Use Case ID	UC-11
Use Case Name	View Destination Details
Actor(s)	Tourist
Preconditions	Tourist has searched for destinations
Postconditions – Success	Destination details are displayed
Postconditions – Failure	Error message displayed if destination unavailable
Main Flow	1. Tourist selects a destination. 2. System retrieves destination details. 3. System displays description, images, pricing, and reviews.
Alternative Flow 1	At step 2: If destination no longer exists, system displays unavailable message.
Alternative Flow 2	Tourist returns to search results page.
Business Rules	BR-12: Only approved destinations are visible to tourists.

Use Case ID	UC-12
Use Case Name	Create Trip Plan
Actor(s)	Tourist
Preconditions	Tourist is logged into the system
Postconditions – Success	Trip plan is saved successfully
Postconditions – Failure	Trip plan is not saved; error displayed
Main Flow	1. Tourist selects destinations and services. 2. Tourist enters travel dates and preferences. 3. System generates itinerary. 4. Tourist confirms trip plan. 5. System saves trip plan.
Alternative Flow 1	At step 3: If selected services are unavailable, system suggests alternatives.
Alternative Flow 2	Tourist edits itinerary before confirmation.
Business Rules	BR-13: Travel dates must be valid future dates.

Use Case ID	UC-13
Use Case Name	Book Hotel
Actor(s)	Tourist, Hotel Owner
Preconditions	Tourist is logged in and hotel rooms are available
Postconditions – Success	Hotel booking is confirmed
Postconditions – Failure	Booking is rejected or unavailable
Main Flow	1. Tourist selects hotel. 2. Tourist chooses room and dates. 3. System checks availability. 4. Tourist confirms booking. 5. System records booking and sends confirmation.
Alternative Flow 1	At step 3: If rooms are unavailable, system displays alternative options.
Alternative Flow 2	Tourist cancels booking before confirmation.
Business Rules	BR-14: Double booking of rooms is not allowed.

Use Case ID	UC-14
Use Case Name	Book Transport
Actor(s)	Tourist, Transport Provider
Preconditions	Tourist is logged in and transport is available
Postconditions – Success	Transport booking is confirmed
Postconditions – Failure	Booking fails due to unavailability
Main Flow	1. Tourist selects transport service. 2. Tourist enters travel details. 3. System checks availability. 4. Tourist confirms booking. 5. System records booking and sends notification.
Alternative Flow 1	At step 3: If no vehicles are available, system displays message.
Alternative Flow 2	Tourist modifies travel details and retries booking.
Business Rules	BR-15: Booking must match provider availability schedule.

Use Case ID	UC-15
Use Case Name	Hire Tourist Guide
Actor(s)	Tourist, Tourist Guide
Preconditions	Tourist guide is available
Postconditions – Success	Guide booking is confirmed
Postconditions – Failure	Guide booking request is rejected
Main Flow	1. Tourist browses available guides. 2. Tourist selects a guide. 3. Tourist submits booking request. 4. Guide accepts request. 5. System confirms booking.
Alternative Flow 1	At step 4: If guide rejects request, system suggests other guides.
Alternative Flow 2	Tourist cancels request before acceptance.
Business Rules	BR-16: Guides can only accept bookings during available time slots.

Use Case ID	UC-16
Use Case Name	Make Payment
Actor(s)	Tourist, Payment Gateway
Preconditions	Tourist has an active booking
Postconditions – Success	Payment is processed successfully
Postconditions – Failure	Payment transaction fails
Main Flow	1. Tourist selects payment option. 2. Tourist enters payment details. 3. System forwards request to payment gateway. 4. Gateway processes payment. 5. System confirms successful payment.
Alternative Flow 1	At step 4: If payment is declined, system displays failure message.
Alternative Flow 2	Tourist cancels payment before submission.
Business Rules	BR-17: Payments must use secure encrypted transactions.

Use Case ID	UC-17
Use Case Name	Cancel Booking
Actor(s)	Tourist
Preconditions	Tourist has an active booking
Postconditions – Success	Booking is cancelled successfully
Postconditions – Failure	Booking remains active
Main Flow	1. Tourist views booking history. 2. Tourist selects booking to cancel. 3. System checks cancellation policy. 4. Tourist confirms cancellation. 5. System cancels booking and updates status.
Alternative Flow 1	At step 3: If cancellation deadline passed, system denies cancellation.
Alternative Flow 2	Refund request is generated automatically if applicable.
Business Rules	BR-18: Cancellation policy depends on provider rules.

Use Case ID	UC-18
Use Case Name	View Booking History
Actor(s)	Tourist
Preconditions	Tourist is logged into the system
Postconditions – Success	Booking history is displayed
Postconditions – Failure	Error message displayed
Main Flow	1. Tourist selects booking history option. 2. System retrieves past and current bookings. 3. System displays booking records.
Alternative Flow 1	At step 2: If no bookings exist, system displays empty history message.
Alternative Flow 2	Tourist filters booking history by date or type.
Business Rules	BR-19: Booking records must be stored for audit purposes.

Use Case ID	UC-19
Use Case Name	Write Reviews
Actor(s)	Tourist
Preconditions	Tourist has completed a booking
Postconditions – Success	Review is published successfully
Postconditions – Failure	Review submission fails
Main Flow	1. Tourist selects completed booking. 2. Tourist writes review and rating. 3. Tourist submits review. 4. System validates and publishes review.
Alternative Flow 1	At step 4: If review contains prohibited content, system rejects submission.
Alternative Flow 2	Tourist edits review before final submission.
Business Rules	BR-20: Only verified tourists can submit reviews.

Use Case ID	UC-20
Use Case Name	Receive Notifications
Actor(s)	Tourist, Notification Center

Preconditions	Tourist has enabled notifications
Postconditions – Success	Notification is delivered successfully
Postconditions – Failure	Notification delivery fails
Main Flow	1. System generates notification. 2. Notification center sends message. 3. Tourist receives notification.
Alternative Flow 1	At step 2: If delivery fails, system retries sending notification.
Alternative Flow 2	Tourist disables notifications in settings.
Business Rules	BR-21: Notifications must be sent for booking updates and payments.

Use Case ID	UC-21
Use Case Name	Request Refund
Actor(s)	Tourist, Payment Gateway
Preconditions	Tourist has cancelled a booking eligible for refund
Postconditions – Success	Refund request is submitted and processed
Postconditions – Failure	Refund request is rejected
Main Flow	1. Tourist selects cancelled booking. 2. Tourist requests refund. 3. System verifies eligibility. 4. Refund request is forwarded to payment gateway. 5. System updates refund status.
Alternative Flow 1	At step 3: If booking is not eligible, system displays rejection message.
Alternative Flow 2	Refund processing is delayed; system notifies tourist.
Business Rules	BR-22: Refund amount depends on cancellation policy.

Use Case ID	UC-22
Use Case Name	Add Hotel / Property
Actor(s)	Hotel Owner
Preconditions	Hotel owner is logged into the system

Postconditions – Success	Hotel/property is added successfully
Postconditions – Failure	Property is not added
Main Flow	1. Hotel owner selects add property option. 2. Enters property details. 3. Uploads images and facilities. 4. Submits property information. 5. System validates and saves property details.
Alternative Flow 1	At step 4: If mandatory fields are missing, system displays validation errors.
Alternative Flow 2	Hotel owner cancels property registration before submission.
Business Rules	BR-23: Each property must have a unique registration ID.

Use Case ID	UC-23
Use Case Name	Manage Rooms
Actor(s)	Hotel Owner
Preconditions	Property exists in the system
Postconditions – Success	Room information is updated successfully
Postconditions – Failure	Room information remains unchanged
Main Flow	1. Hotel owner selects property. 2. Opens room management section. 3. Adds, edits, or removes room details. 4. System validates information. 5. Room data is updated.
Alternative Flow 1	At step 4: If room capacity is invalid, system displays error.
Alternative Flow 2	Hotel owner cancels changes before saving.
Business Rules	BR-24: Room details must include capacity and pricing information.

Use Case ID	UC-24
Use Case Name	Set Room Prices
Actor(s)	Hotel Owner
Preconditions	Rooms are registered in the system

Postconditions – Success	Room prices are updated successfully
Postconditions – Failure	Pricing update fails
Main Flow	1. Hotel owner selects room type. 2. Enters pricing details. 3. System validates pricing information. 4. Updated prices are saved.
Alternative Flow 1	At step 3: If price value is invalid, system displays error message.
Alternative Flow 2	Hotel owner schedules seasonal pricing updates.
Business Rules	BR-25: Room prices must be greater than zero.

Use Case ID	UC-25
Use Case Name	View Revenue Reports
Actor(s)	Hotel Owner
Preconditions	Hotel owner has completed financial transactions
Postconditions – Success	Revenue reports are generated successfully
Postconditions – Failure	Report generation fails
Main Flow	1. Hotel owner selects revenue report option. 2. Chooses reporting period. 3. System retrieves transaction records. 4. System generates revenue report. 5. Report is displayed.
Alternative Flow 1	At step 3: If no revenue data exists, system displays empty report message.
Alternative Flow 2	Hotel owner exports report for offline use.
Business Rules	BR-26: Revenue calculations must include only completed transactions.

Use Case ID	UC-26
Use Case Name	Add Vehicles
Actor(s)	Transport Provider
Preconditions	Provider is logged into the system

Postconditions – Success	Vehicle is added successfully
Postconditions – Failure	Vehicle registration fails
Main Flow	1. Provider selects add vehicle option. 2. Enters vehicle details. 3. Uploads required documents. 4. Submits vehicle information. 5. System validates and saves vehicle details.
Alternative Flow 1	At step 4: If mandatory fields are missing, system displays validation errors.
Alternative Flow 2	Provider cancels vehicle registration before submission.
Business Rules	BR-27: Each vehicle must have a unique registration number.

Use Case ID	UC-27
Use Case Name	Update Vehicle Details
Actor(s)	Transport Provider
Preconditions	Vehicle exists in the system
Postconditions – Success	Vehicle details are updated successfully
Postconditions – Failure	Vehicle details remain unchanged
Main Flow	1. Provider selects vehicle. 2. Opens edit vehicle details option. 3. Updates vehicle information. 4. System validates changes. 5. Updated details are saved successfully.
Alternative Flow 1	At step 4: If invalid data is entered, system displays error message.
Alternative Flow 2	Provider cancels modifications before saving.
Business Rules	BR-28: Vehicle capacity and license information must be valid.

Use Case ID	UC-28
Use Case Name	Set Pricing
Actor(s)	Transport Provider
Preconditions	Vehicles are registered in the system

Postconditions – Success	Pricing details are updated successfully
Postconditions – Failure	Pricing update fails
Main Flow	1. Provider selects vehicle or service type. 2. Enters pricing details. 3. System validates price information. 4. Updated pricing is saved successfully.
Alternative Flow 1	At step 3: If price is invalid, system displays error message.
Alternative Flow 2	Provider schedules seasonal price updates.
Business Rules	BR-29: Service price must be greater than zero.

Use Case ID	UC-29
Use Case Name	Track Trips
Actor(s)	Transport Provider
Preconditions	Active trips exist in the system
Postconditions – Success	Trip status and location are displayed
Postconditions – Failure	Trip tracking information unavailable
Main Flow	1. Provider opens trip tracking section. 2. System retrieves active trip data. 3. Current trip status and route are displayed. 4. Provider monitors ongoing trips.
Alternative Flow 1	At step 2: If GPS/location data is unavailable, system displays tracking error.
Alternative Flow 2	Provider refreshes trip status manually.
Business Rules	BR-30: Trip updates must refresh in real time whenever possible.

Use Case ID	UC-30
Use Case Name	Add Tour Services
Actor(s)	Tourist Guide
Preconditions	Tourist guide is logged into the system

Postconditions – Success	Tour service is added successfully
Postconditions – Failure	Tour service creation fails
Main Flow	1. Tourist guide selects add service option. 2. Enters tour service details. 3. Uploads images and descriptions. 4. Submits service information. 5. System validates and saves the service.
Alternative Flow 1	At step 4: If mandatory fields are missing, system displays validation errors.
Alternative Flow 2	Tourist guide cancels service creation before submission.
Business Rules	BR-31: Each tour service must have a unique identifier.

Use Case ID	UC-31
Use Case Name	Update Tour Details
Actor(s)	Tourist Guide
Preconditions	Tour service exists in the system
Postconditions – Success	Tour details are updated successfully
Postconditions – Failure	Tour details remain unchanged
Main Flow	1. Tourist guide selects existing tour service. 2. Opens edit option. 3. Updates service information. 4. System validates changes. 5. Updated details are saved successfully.
Alternative Flow 1	At step 4: If invalid data is entered, system displays validation errors.
Alternative Flow 2	Tourist guide cancels modifications before saving.
Business Rules	BR-32: Tour pricing and schedule must be valid.

Use Case ID	UC-32
Use Case Name	Manage Tours
Actor(s)	Tourist Guide
Preconditions	Tour services are registered in the system

Postconditions – Success	Tour information is managed successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Tourist guide opens manage tours section. 2. System displays available tours. 3. Guide adds, edits, or removes tours. 4. System validates updates. 5. Changes are saved successfully.
Alternative Flow 1	At step 4: If scheduling conflicts exist, system rejects changes.
Alternative Flow 2	Tourist guide cancels modifications before saving.
Business Rules	BR-33: Tour schedules must not overlap with confirmed bookings.

Use Case ID	UC-33
Use Case Name	Provide Guidance Service
Actor(s)	Tourist Guide, Tourist
Preconditions	Booking is confirmed and tour date has arrived
Postconditions – Success	Guidance service is completed successfully
Postconditions – Failure	Service is interrupted or cancelled
Main Flow	1. Tourist guide reviews scheduled tour. 2. Meets tourist at designated location. 3. Conducts guided tour activities. 4. Provides destination information and assistance. 5. Marks service as completed in the system.
Alternative Flow 1	At step 2: If tourist does not arrive, system records no-show status.
Alternative Flow 2	Tour is cancelled due to weather or emergency conditions.
Business Rules	BR-34: Guides must provide services only for confirmed bookings.

Use Case ID	UC-34
Use Case Name	Admin Login
Actor(s)	Admin
Preconditions	Admin account exists in the system

Postconditions – Success	Admin is authenticated and dashboard is displayed
Postconditions – Failure	Login fails; error message displayed
Main Flow	1. Admin opens login page. 2. Enters username and password. 3. System validates credentials. 4. System authenticates admin. 5. Admin dashboard is displayed.
Alternative Flow 1	At step 3: If credentials are invalid, system displays error message.
Alternative Flow 2	At step 2: If fields are empty, system requests required information.
Business Rules	BR-35: Only authorized administrators can access admin functions.

Use Case ID	UC-35
Use Case Name	Manage Users
Actor(s)	Admin
Preconditions	Admin is logged into the system
Postconditions – Success	User records are updated successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens user management section. 2. System displays registered users. 3. Admin searches or selects a user. 4. Admin updates, activates, or deactivates user account. 5. System saves changes.
Alternative Flow 1	At step 3: If user does not exist, system displays error message.
Alternative Flow 2	Admin cancels modifications before saving.
Business Rules	BR-36: Admin can suspend accounts violating platform policies.

Use Case ID	UC-36
Use Case Name	Manage Destinations
Actor(s)	Admin
Preconditions	Admin is authenticated

Postconditions – Success	Destination information is updated successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens destination management section. 2. Views destination list. 3. Adds, edits, or removes destination details. 4. System validates updates. 5. System saves destination information.
Alternative Flow 1	At step 4: If mandatory fields are missing, system displays validation errors.
Alternative Flow 2	Admin cancels changes before saving.
Business Rules	BR-37: Only approved destinations can be published to tourists.

Use Case ID	UC-37
Use Case Name	Manage Hotels
Actor(s)	Admin
Preconditions	Hotel properties exist in the system
Postconditions – Success	Hotel records are managed successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens hotel management section. 2. System displays hotel records. 3. Admin reviews hotel details. 4. Admin approves, updates, or removes hotel information. 5. System saves updates.
Alternative Flow 1	At step 3: If hotel details are invalid, admin rejects registration.
Alternative Flow 2	Admin cancels action before confirmation.
Business Rules	BR-38: Hotels must meet platform verification requirements before approval.

Use Case ID	UC-38
Use Case Name	Manage Transport Providers
Actor(s)	Admin

Preconditions	Transport providers are registered in the system
Postconditions – Success	Provider information is updated successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens transport provider management section. 2. System displays provider records. 3. Admin reviews provider details. 4. Admin approves, updates, or removes provider information. 5. System saves updates.
Alternative Flow 1	At step 3: If provider documents are invalid, registration is rejected.
Alternative Flow 2	Admin suspends provider account for policy violations.
Business Rules	BR-39: Providers must submit valid license information.

Use Case ID	UC-39
Use Case Name	Manage Tourist Guides
Actor(s)	Admin
Preconditions	Tourist guides are registered in the system
Postconditions – Success	Guide information is updated successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens tourist guide management section. 2. System displays guide records. 3. Admin reviews guide details and certifications. 4. Admin approves, updates, or removes guide profiles. 5. System saves changes.
Alternative Flow 1	At step 3: If certifications are invalid, registration is rejected.
Alternative Flow 2	Admin suspends guide account due to complaints or violations.
Business Rules	BR-40: Tourist guides must provide valid certification documents.

Use Case ID	UC-40
Use Case Name	Monitor Bookings
Actor(s)	Admin
Preconditions	Booking records exist in the system

Postconditions – Success	Booking information is monitored successfully
Postconditions – Failure	Booking data cannot be retrieved
Main Flow	1. Admin opens booking monitoring section. 2. System retrieves booking records. 3. Admin reviews booking activities and statuses. 4. System displays booking analytics.
Alternative Flow 1	At step 2: If records are unavailable, system displays error message.
Alternative Flow 2	Admin filters bookings by provider, tourist, or date.
Business Rules	BR-41: All booking activities must be logged for auditing purposes.

Use Case ID	UC-41
Use Case Name	Manage Payments
Actor(s)	Admin, Payment Gateway
Preconditions	Payment transactions exist in the system
Postconditions – Success	Payment records are updated and verified
Postconditions – Failure	Payment update fails
Main Flow	1. Admin opens payment management section. 2. System retrieves transaction records. 3. Admin reviews payment status. 4. Admin verifies or resolves payment issues. 5. System updates payment records.
Alternative Flow 1	At step 3: If transaction fails, admin marks payment as disputed or pending.
Alternative Flow 2	Admin processes refund requests manually when required.
Business Rules	BR-42: Payment records must be securely stored and traceable.

Use Case ID	UC-42
Use Case Name	Generate Reports
Actor(s)	Admin
Preconditions	System contains operational data

Postconditions – Success	Reports are generated successfully
Postconditions – Failure	Report generation fails
Main Flow	1. Admin selects report generation option. 2. Chooses report type and period. 3. System retrieves required data. 4. System generates report. 5. Report is displayed or exported.
Alternative Flow 1	At step 3: If data is unavailable, system displays empty report message.
Alternative Flow 2	Admin exports report in downloadable format.
Business Rules	BR-43: Reports must contain accurate and up-to-date system data.

Use Case ID	UC-43
Use Case Name	Manage Reviews
Actor(s)	Admin
Preconditions	Reviews exist in the system
Postconditions – Success	Reviews are moderated successfully
Postconditions – Failure	Review moderation action fails
Main Flow	1. Admin opens review management section. 2. System displays submitted reviews. 3. Admin reviews reported or flagged content. 4. Admin approves or removes reviews. 5. System updates review status.
Alternative Flow 1	At step 3: If review violates policies, system removes it automatically after confirmation.
Alternative Flow 2	Admin restores previously removed review after appeal review.
Business Rules	BR-44: Offensive or misleading reviews must be removed.

Use Case ID	UC-44
Use Case Name	Handle Complaints
Actor(s)	Admin
Preconditions	Complaints have been submitted by users

Postconditions – Success	Complaint is resolved and updated
Postconditions – Failure	Complaint remains unresolved
Main Flow	1. Admin opens complaint management section. 2. System displays submitted complaints. 3. Admin reviews complaint details. 4. Admin investigates issue. 5. Admin resolves complaint and updates status.
Alternative Flow 1	At step 4: If additional evidence is needed, admin requests more information from users.
Alternative Flow 2	Complaint is escalated to higher authority for further review.
Business Rules	BR-45: Complaints must be resolved within the defined response period.

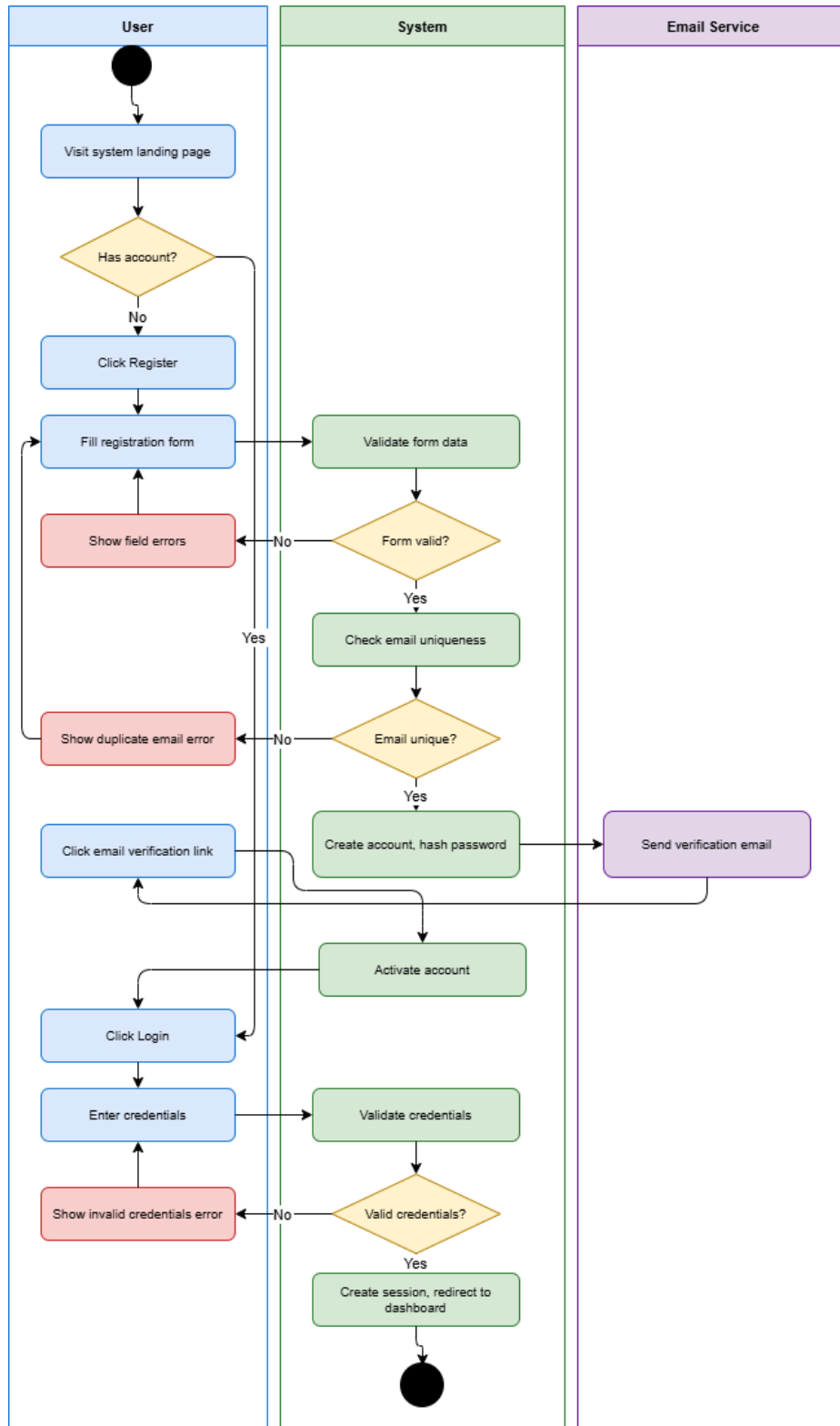
Use Case ID	UC-45
Use Case Name	Manage Notifications
Actor(s)	Admin, Notification Center
Preconditions	Notification service is active
Postconditions – Success	Notifications are created and delivered successfully
Postconditions – Failure	Notification delivery fails
Main Flow	1. Admin opens notification management section. 2. Creates or schedules notification. 3. System validates notification details. 4. Notification center sends messages to users. 5. System logs delivery status.
Alternative Flow 1	At step 4: If delivery fails, system retries notification delivery.
Alternative Flow 2	Admin cancels scheduled notification before sending.
Business Rules	BR-46: Critical notifications must be prioritized for immediate delivery.

Use Case ID	UC-46
Use Case Name	System Maintenance
Actor(s)	Admin

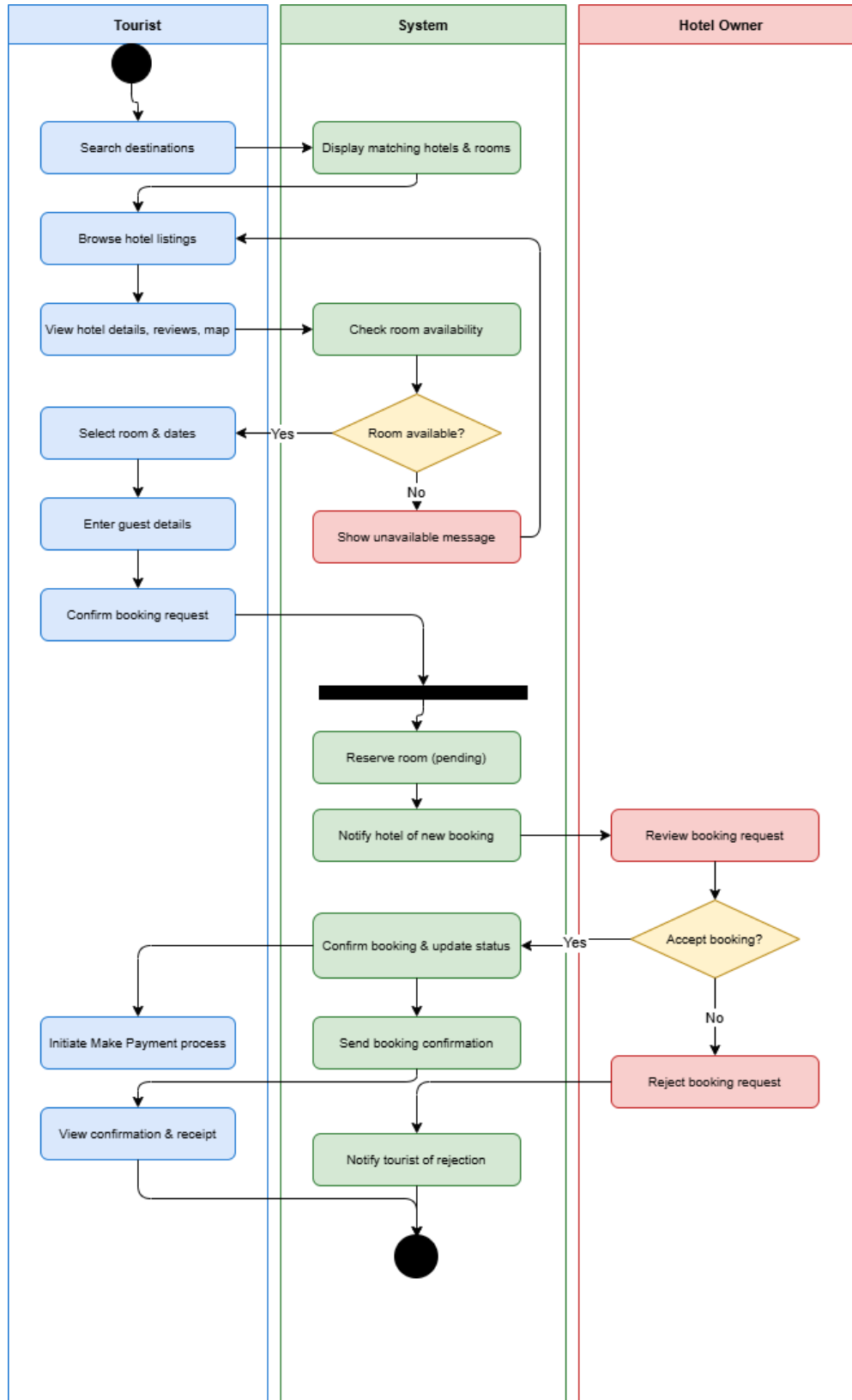
Preconditions	Admin has maintenance privileges
Postconditions – Success	System maintenance tasks are completed successfully
Postconditions – Failure	Maintenance task fails or system remains unstable
Main Flow	1. Admin opens maintenance panel. 2. Selects maintenance task (backup, update, cleanup, etc.). 3. System performs maintenance operation. 4. System displays completion status.
Alternative Flow 1	At step 3: If maintenance operation fails, system logs error details.
Alternative Flow 2	Admin schedules maintenance for a later time.
Business Rules	BR-47: Regular backups must be performed before major system updates.

2.5 Activity diagrams

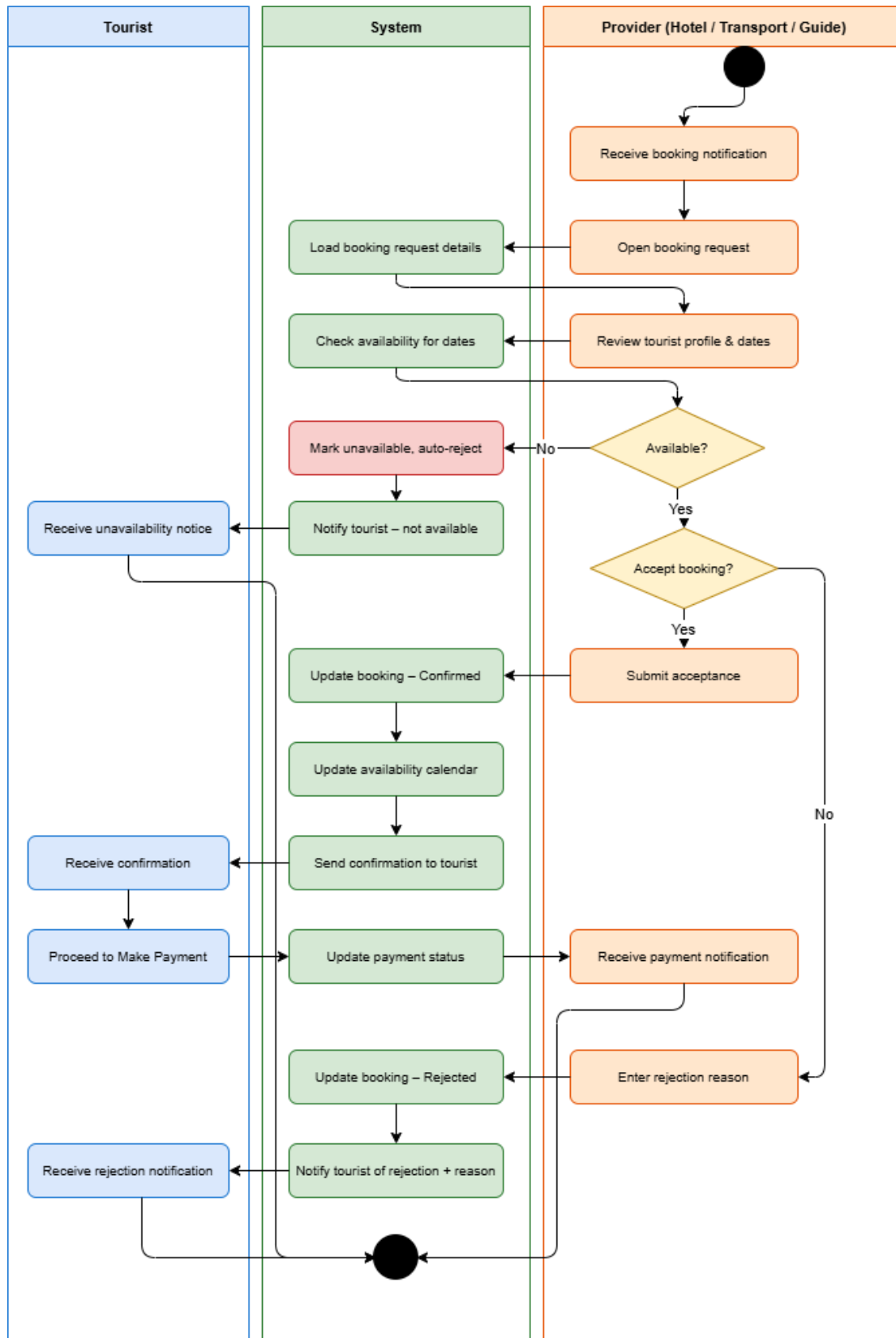
UC-01: Register/Login



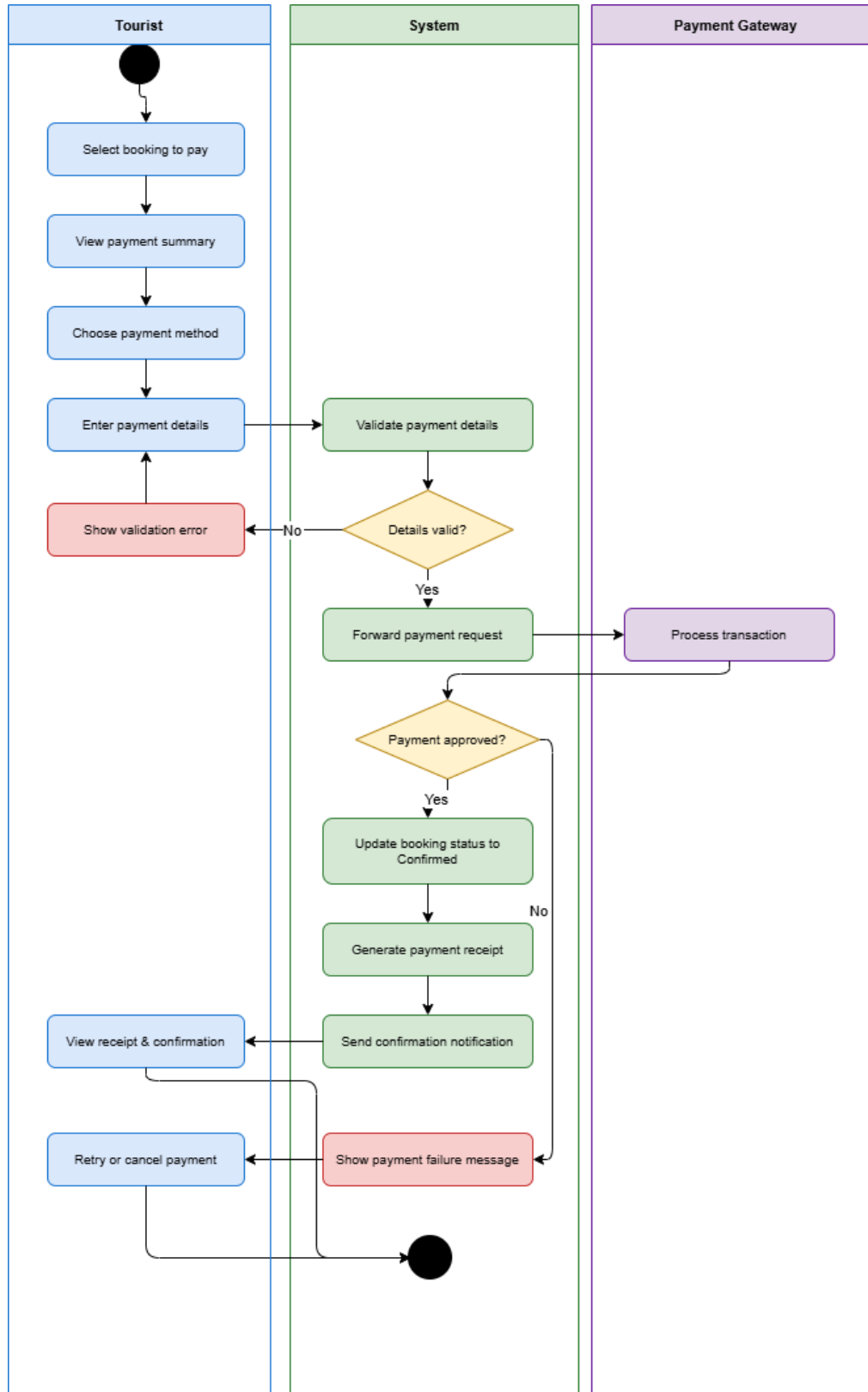
UC-13: Book Hotel



UC-04: Accept/ Reject Bookings



UC-16: Make Payment



UC-17: Cancel Booking & UC-21: Refund

